Nitroprusside: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Nitroprusside: Patient drug information" and "Nitroprusside: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Appropriate administration:

Nitropress: After reconstitution, nitroprusside is not suitable for direct injection. The reconstituted solution must be further diluted in dextrose 5% injection before infusion.

Hypotension:

Nitroprusside can cause precipitous decreases in blood pressure. In patients not properly monitored, these decreases can lead to irreversible ischemic injuries or death. Use only when available equipment and personnel allow blood pressure to be continuously monitored.

Cyanide toxicity:

Except when used briefly or at low (less than 2 mcg/kg/min) infusion rates, nitroprusside injection gives rise to important quantities of cyanide ion, which can reach toxic, potentially lethal levels. The usual dose rate is 0.5 to 10 mcg/kg/min, but infusion at the maximum dose rates should never last more than 10 minutes. If blood pressure has not been adequately controlled after 10 minutes of infusion at the maximum rate, terminate administration immediately. Although acid-base balance and venous oxygen concentration should be monitored and may indicate cyanide toxicity, these laboratory tests provide imperfect guidance.

Brand Names: US

- Nipride RTU

Brand Names: Canada

- Nipride

Pharmacologic Category

- Antihypertensive;
- Vasodilator

Dosing: Adult

Note: Safety: Use may be limited due to risk for cyanide and thiocyanate toxicity, which is increased with higher doses (eg, >2 mcg/kg/minute) and/or prolonged duration of use, especially in patients with kidney or hepatic impairment. Invasive blood pressure monitoring, preferably via arterial line in a critical care unit, is recommended for appropriate dose titration.

Acute aortic syndromes/acute aortic dissection

Acute aortic syndromes/acute aortic dissection (adjunctive agent) (off-label use):

Note: Manage patients on an emergency basis (including operative assessment) by first controlling pain with IV opioids and heart rate (target 60 to 80 beats per minute) with a parenteral beta blocker (eg, esmolol) while targeting systolic BP of <120 mm Hg (or lowest tolerated pressure without compromising perfusion). If systolic BP remains elevated after heart rate is controlled at 60 to 80 beats per minute with beta-blockade, may consider adding a vasodilator (eg, nitroprusside) (Ref).

Continuous infusion: IV: Initial: 0.25 to 0.5 mcg/kg/minute; titrate as needed by 0.5 mcg/kg/minute every 5 minutes to achieve desired hemodynamic effect; maximum dose: 10 mcg/kg/minute. Higher doses (eg, 8 to 10 mcg/kg/minute) should only be used for a maximum of 10 minutes; use for the shortest duration possible to avoid toxicity (Ref).

Acute decompensated heart failure

Acute decompensated heart failure (adjunctive agent):

Note: May consider in patients with volume overload in the absence of symptomatic hypotension to help relieve congestion and dyspnea as an adjunct to IV diuretics (Ref).

Continuous infusion: IV: Initial: 0.1 to 0.3 mcg/kg/minute; titrate as needed every 5 to 15 minutes to achieve desired hemodynamic effect; usual dosage range: 1 to 3 mcg/kg/minute; maximum dose: 5 mcg/kg/minute for an 80 kg patient (Ref).

Acute ischemic stroke, BP management with reperfusion therapy

Acute ischemic stroke, BP management with reperfusion therapy (off-label use):

Note: During and 24 hours after reperfusion therapy, maintain a target BP of **≤180/105 mm Hg**; reserve nitroprusside for hypertension that is refractory to first-line agents or diastolic BP >140 mm Hg (Ref).

Continuous infusion: IV: Initial: 0.25 to 0.5 mcg/kg/minute; titrate by 0.5 mcg/kg/minute every 5 minutes. Maximum dose: 10 mcg/kg/minute (for a maximum of 10 minutes); consider a maximum dose of 2 mcg/kg/minute to avoid toxicity (Ref).

Hypertensive emergency

Hypertensive emergency:

Note: Consider for use when preferred agents with less toxicity are unavailable. When used for hypertensive emergency without a compelling condition, in general, reduce mean arterial BP gradually by ~10% to 20% over the first hour. If stable, may then reduce to a goal BP of <160/100 mm Hg for the next 2 to 6 hours, then to 130 to 140 mm Hg over the next 24 to 48 hours to limit target organ injury. Invasive BP monitoring is recommended for appropriate dose titration (Ref).

Continuous infusion: IV: Initial: 0.25 to 0.5 mcg/kg/minute; titrate as needed by 0.5 mcg/kg/minute every 5 minutes to achieve the target blood pressure; to avoid toxicity, limit dose to ≤2 mcg/kg/minute if possible; maximum dose: 10 mcg/kg/minute. Higher doses (eg, 8 to 10 mcg/kg/minute) should only be used for a maximum of 10 minutes; use for the shortest duration possible to avoid toxicity (Ref).

Dosing: Kidney Impairment: Adult

Use in patients with kidney impairment may lead to the accumulation of thiocyanate and subsequent toxicity; limit use.

eGFR <30 mL/minute/1.73 m^2: **IV:** Limit mean infusion rate to <3 mcg/kg/minute.

Anuric patients: **IV:** Limit mean infusion rate to 1 mcg/kg/minute.

Dosing: Liver Impairment: Adult

No dosage adjustment provided in manufacturer's labeling; due to the risk of cyanide toxicity, use with caution.

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see ”Nitroprusside: Pediatric drug information”)

Note: Increased infusion rates and prolonged use have been associated with increased cyanide concentrations and cyanide toxicity; monitor cyanide levels with prolonged use (eg, >72 hours); coadministration with sodium thiosulfate is recommended (Ref):

Hypertension, acute including hypertensive crisis

Hypertension, acute including hypertensive crisis: Infants, Children, and Adolescents: Continuous IV infusion: Initial: 0.3 to 0.5 mcg/kg/minute; titrate every 5 minutes to desired effect; usual dose: 3 mcg/kg/minute; maximum dose: 10 mcg/kg/minute (Ref).

Cardiac output maintenance/stabilization, post resuscitation

Cardiac output maintenance/stabilization, post resuscitation: Limited data available:

Infants, Children, and Adolescents: Continuous IV infusion: Initial: 0.3 to 1 mcg/kg/minute; titrate to desired effect; maximum dose: 8 mcg/kg/minute (Ref).

Dosing: Kidney Impairment: Pediatric

Infants, Children, and Adolescents: Use in patients with renal impairment may lead to the accumulation of thiocyanate and subsequent toxicity; limit use and monitor thiocyanate blood concentrations:

eGFR <30 mL/minute/1.73 m^2: Limit mean infusion rate to <3 mcg/kg/minute.

Anuric patients: Limit mean infusion rate to 1 mcg/kg/minute.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling; due to the risk of cyanide toxicity, use with caution.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Frequency not defined.

Cardiovascular: Bradycardia, ECG changes, flushing, palpitations, severe hypotension, substernal pain, tachycardia

Central nervous system: Apprehension, dizziness, headache, increased intracranial pressure, restlessness

Dermatologic: Diaphoresis, localized erythematous streaking, skin rash

Endocrine & metabolic: Hypothyroidism

Gastrointestinal: Abdominal pain, intestinal obstruction, nausea, retching

Hematologic & oncologic: Decreased platelet aggregation, methemoglobinemia

Local: Irritation at injection site

Neuromuscular & skeletal: Muscle twitching

Contraindications

Treatment of compensatory hypertension (aortic coarctation, arteriovenous shunting); to produce controlled hypotension during surgery in patients with known inadequate cerebral circulation or in moribund patients (A.S.A. Class 5E) requiring emergency surgery; acute heart failure associated with reduced systemic vascular resistance (eg, septic shock); congenital (Leber's) optic atrophy or tobacco amblyopia; concomitant use with avanafil, sildenafil, tadalafil, vardenafil, or riociguat

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindication (not in US labeling): Hypersensitivity to nitroprusside or any component of the formulation; uncorrected anemia or hypovolemia; hepatic disease; severe kidney disease; disease states associated with vitamin B_{12} deficiency

Warnings/Precautions

Concerns related to adverse effects:

• Cyanide toxicity: **[US Boxed Warning]: Except when used briefly or at low (<2 mcg/kg/minute) infusion rates, nitroprusside gives rise to large cyanide quantities. Do not use the maximum dose for more than 10 minutes; if blood pressure is not controlled by the maximum rate (ie, 10 mcg/kg/minute) after 10 minutes, discontinue infusion. Monitor for cyanide toxicity via acid-base balance and venous oxygen concentration; however, clinicians should note that these indicators may not always reliably indicate cyanide toxicity.** Patients at risk of cyanide toxicity include those who are malnourished, have hepatic impairment, or those undergoing cardiopulmonary bypass, or therapeutic hypothermia (Rindone 1992). Discontinue use of nitroprusside if signs and/or symptoms of cyanide toxicity (eg, metabolic acidosis, decreased oxygen saturation, bradycardia, confusion, convulsions) occur. Although not routinely done, sodium thiosulfate has been co-administered with nitroprusside using a 10:1 ratio of sodium thiosulfate to nitroprusside when higher doses of nitroprusside are used (eg, 4 to 10 mcg/kg/minute) for extended periods of time in order to prevent cyanide toxicity (Shulz 2010; Varon 2008); thiocyanate toxicity may still occur with this approach (Rindone 1992). The use of other agents (eg, clevidipine, labetalol, nicardipine) should be considered if blood pressure is not controlled with nitroprusside.

• Hypotension: **[US Boxed Warning]: Excessive hypotension resulting in compromised perfusion of vital organs may occur; continuous blood pressure monitoring by experienced personnel is required.**

• Increased intracranial pressure: Use may elevate intracranial pressure; in patients whose intracranial pressure is already elevated, use only with extreme caution.

Methemoglobinemia: Nitroprusside can cause a dose-dependent conversion of hemoglobin to methemoglobin. Methemoglobinemia should be suspected in any patient receiving >10 mg/kg of nitroprusside and exhibiting signs of impaired oxygen delivery despite adequate cardiac output and arterial pO_2. Symptomatic patients, regardless of methemoglobin level should be treated with methylene blue (first-line). Treatment is suggested if level is ≥30% even if asymptomatic unless patient has a preexisting condition (eg, coronary artery disease) and are incapable of tolerating reductions in oxygen carrying capacity then treatment is suggested if level reaches 10% (Cortazzo 2013).

• Thiocyanate toxicity: Can occur in patients with kidney impairment or those on prolonged infusions (ie, >3 mcg/kg/minute for >72 hours).

Disease-related concerns:

• Anemia: When nitroprusside is used for controlled hypotension during surgery, correct pre-existing anemia prior to use when possible.

• Hepatic impairment: Use with extreme caution in patients with hepatic impairment.

• Hypovolemia: When nitroprusside is used for controlled hypotension during surgery, correct pre-existing hypovolemia prior to use when possible.

• Kidney impairment: Use with extreme caution in patients with kidney impairment; use the lowest end of the dosage range; monitor thiocyanate concentrations closely.

• Myocardial infarction: Use caution in patients with acute myocardial infarction because of hemodynamic effects and possible coronary steal.

Other warnings/precautions:

• Appropriate administration: Nitropress: **[US Boxed Warning]: Solution must be further diluted with 5% dextrose in water. Do not administer by direct injection.**

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as sodium:

Nipride RTU: 20 mg/100 mL in NaCl 0.9% (100 mL)

Generic: 25 mg/mL (2 mL); 50 mg/2 mL (2 mL)

Solution, Intravenous, as sodium [preservative free]:

Nipride RTU: 50 mg/100 mL in NaCl 0.9% (100 mL)

Generic: 25 mg/mL (2 mL); 20 mg/100 mL in NaCl 0.9% (100 mL); 50 mg/100 mL in NaCl 0.9% (100 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Nipride RTU Intravenous)

20 mg/100 mL 0.9% (per mL): $0.54

50 mg/100 mL 0.9% (per mL): $0.66

Solution (Nitroprusside Sodium Intravenous)

25 mg/mL (per mL): $9.75 - $161.82

Solution (Nitroprusside Sodium-NaCl Intravenous)

20 mg/100 mL 0.9% (per mL): $0.43

50 mg/100 mL 0.9% (per mL): $0.53

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous, as sodium:

Nipride: 25 mg/mL (2 mL)

Generic: 25 mg/mL (2 mL)

Administration: Adult

IV: IV infusion only; infusion pump required. Due to potential for excessive hypotension, continuously monitor patient's blood pressure during therapy. Product should always be protected from light, even during administration.

Preparation for Administration: Adult

Nitropress: Prior to administration, nitroprusside sodium should be further diluted by diluting 50 mg in 250 to 1,000 mL of D5W (preferred), LR, or NS.

Nipride RTU vial: Premixed solutions are available.

Use only clear solutions; solutions of nitroprusside exhibit a color described as brownish, brown, brownish-pink, light orange, and straw. Solutions are highly sensitive to light. Exposure to light causes decomposition, resulting in a highly colored solution of orange, dark brown or blue. **A blue color indicates almost complete decomposition.** Do not use discolored solutions (eg, blue, green, red) or solutions in which particulate matter is visible.

Prepared solutions should be wrapped as soon as possible with aluminum foil or other opaque material to protect from light.

Usual Infusion Concentrations: Adult

IV infusion: 50 mg in 250 mL (concentration: 200 **mcg**/mL) **or** 100 mg in 250 mL (concentration: 400 **mcg**/mL) of D_5W

Administration: Pediatric

Parenteral: Administer as a continuous IV infusion via infusion pump. Solution should be protected from light, but not necessary to wrap administration set or IV tubing.

Concentrated solution (25 **mg**/mL): Not for direct injection; **must dilute** prior to administration.

Premixed solutions (200 mcg/mL, 500 mcg/mL): No further dilution required.

Preparation for Administration: Pediatric

Parenteral: Solutions may range from clear/colorless to red/brown in color. Do not use solutions with particulate matter or those that are discolored (eg, blue, green, bright red). Discoloration is a sign of decomposition; exposure to light causes decomposition, resulting in a highly colored solution of bright red, green, or blue. A blue color indicates almost complete decomposition. Do not add other medications to nitroprusside solutions. **Prepared solutions should be wrapped with aluminum foil or other opaque material to protect from light (do as soon as possible).**

Concentrated solution (25 **mg**/mL): **Must dilute** prior to administration; dilute with D5W (preferred), LR, or NS to a concentration of 50 to 200 mcg/mL (Ref); ASHP recommends standard concentrations of 200 mcg/mL and 500 mcg/mL in pediatric patients (Ref); in fluid-restricted patients, concentrations up to 1,000 mcg/mL in D5W have been used (Ref).

Usual Infusion Concentrations: Pediatric

Note: Premixed solutions available.

IV infusion: 100 **mcg**/mL **or** 200 **mcg**/mL.

Storage/Stability

Nitropress, Nipride RTU vial: Store at 20°C to 25°C (68°F to 77°F). Protect from light; recommended to store in carton until used.

Stability of parenteral admixture in D5W, LR, or NS at room temperature (25°C) and at refrigeration temperature (4°C) is 24 hours.

Use: Labeled Indications

Acute decompensated heart failure: Management of acute decompensated heart failure.

Hypertensive emergency: Management of hypertensive crises; used for controlled hypotension to reduce bleeding during surgery.

Use: Off-Label: Adult

Acute aortic syndromes/acute aortic dissection; Acute ischemic stroke, BP management with reperfusion therapy

Medication Safety Issues

Sound-alike/look-alike issues:

Nitroprusside may be confused with nitroglycerin

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drugs which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Blood Pressure Lowering Agents: May increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Brigatinib: May decrease antihypertensive effects of Antihypertensive Agents. Brigatinib may increase bradycardic effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Dapsone (Topical): May increase adverse/toxic effects of Methemoglobinemia Associated Agents. *Risk C: Monitor*

Dexmethylphenidate: May decrease therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Flunarizine: May increase therapeutic effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Increasing Effects: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Indoramin: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Isocarboxazid: May increase antihypertensive effects of Antihypertensive Agents. *Risk X: Avoid*

Local Anesthetics: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Local Anesthetics. Specifically, the risk for methemoglobinemia may be increased. *Risk C: Monitor*

Loop Diuretics: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Methylphenidate: May decrease antihypertensive effects of Antihypertensive Agents. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitric Oxide: May increase adverse/toxic effects of Methemoglobinemia Associated Agents. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perazine: May increase hypotensive effects of Antihypertensive Agents. *Risk C: Monitor*

Periciazine: Antihypertensive Agents may increase orthostatic hypotensive effects of Periciazine. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Nitroprusside. *Risk X: Avoid*

Prazosin: Antihypertensive Agents may increase hypotensive effects of Prazosin. *Risk C: Monitor*

Prilocaine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Prilocaine. Combinations of these agents may increase the likelihood of significant methemoglobinemia. Management: Monitor for signs of methemoglobinemia when prilocaine is used in combination with other agents associated with development of methemoglobinemia. Avoid use of these agents with prilocaine/lidocaine cream in infants less than 12 months of age. *Risk C: Monitor*

Primaquine: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Primaquine. Specifically, the risk for methemoglobinemia may be increased. Management: Avoid concomitant use of primaquine and other drugs that are associated with methemoglobinemia when possible. If combined, monitor methemoglobin levels closely. *Risk D: Consider Therapy Modification*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Riociguat: May increase hypotensive effects of Nitroprusside. *Risk X: Avoid*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Sodium Nitrite: Methemoglobinemia Associated Agents may increase adverse/toxic effects of Sodium Nitrite. Combinations of these agents may increase the likelihood of significant methemoglobinemia. *Risk C: Monitor*

Terazosin: Antihypertensive Agents may increase hypotensive effects of Terazosin. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Urapidil: Antihypertensive Agents may increase hypotensive effects of Urapidil. *Risk C: Monitor*

Pregnancy Considerations

Animal studies have shown that nitroprusside may cross the placental barrier and result in fetal cyanide levels that are dose-related to maternal nitroprusside levels. However, information related to use in pregnancy is limited.

Breastfeeding Considerations

It is not known if nitroprusside is present in breast milk. Due to the potential for serious adverse reactions in the breastfed infant, a decision should be made whether to discontinue breastfeeding or to discontinue the drug, taking into account the importance of treatment to the mother.

Monitoring Parameters

Blood pressure via arterial line and heart rate (cardiac monitor and blood pressure monitor required) (Friedrich 1995); monitor for cyanide and thiocyanate toxicity; monitor venous oxygen saturation; monitor acid-base status as acidosis can be the earliest sign of cyanide toxicity; monitor thiocyanate levels if requiring prolonged infusion (>3 days) or dose >3 mcg/kg/minute or patient has kidney dysfunction; monitor cyanide blood levels (if available with appropriate turnaround time) in patients with decreased hepatic function

Consult individual institutional policies and procedures.

Reference Range

Serum thiocyanate levels are not helpful in detecting toxicity. A level may be confirmatory if a patient is exhibiting signs and symptoms of thiocyanate toxicity. Initial signs of toxicity (eg, tinnitus) may be observed at levels >35 mcg/mL (SI: >602 micromole/L) (manufacturer suggests 60 mcg/mL [SI: 1,032 micromole/L]), but serious toxicity typically may not occur with levels <100 mcg/mL (SI: <1,720 micromole/L).

Mechanism of Action

Causes peripheral vasodilation by direct action on venous and arteriolar smooth muscle, thus reducing peripheral resistance; will increase cardiac output by decreasing afterload; reduces aortal and left ventricular impedance

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Hypotensive effect: <2 minutes

Duration: Hypotensive effect: 1-10 minutes

Metabolism: Nitroprusside combines with hemoglobin to produce cyanide and cyanmethemoglobin. Cyanide detoxification occurs via rhodanase-mediated conversion of cyanide to thiocyanate; rhodanase couples cyanide molecules to sulfane sulfur groups from a sulfur donor (eg, thiosulfate, cystine, cysteine). This process has limited capacity and may become overwhelmed with large exposures once sulfur donor supplies are exhausted resulting in toxicity.

Half-life elimination: Nitroprusside, circulatory: ~2 minutes; Thiocyanate, elimination: ~3 days (may be doubled or tripled in renal failure)

Excretion: Urine (as thiocyanate)

Patient Counseling Points

(For additional information see ”Nitroprusside: Patient drug information”)

What is this drug used for?

• It is used to lower blood pressure.

• It is used to treat heart failure (weak heart).

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Acidosis like confusion, fast breathing, fast heartbeat, abnormal heartbeat, severe abdominal pain, nausea, vomiting, fatigue, shortness of breath, or loss of strength and energy.

• Methemoglobinemia like blue or gray color of the lips, nails, or skin; abnormal heartbeat; seizures; severe dizziness or passing out; severe headache; fatigue; loss of strength and energy; or shortness of breath.

• Severe headache

• Small pupils

• Severe dizziness

• Passing out

• Noise or ringing in the ears

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine’s uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Nitropress;
- (AR) Argentina: Niprusodio | Nitroprusiato de sodio gemepe;
- (BR) Brazil: Nitrop;
- (CH) Switzerland: Nipruss;
- (CL) Chile: Nitropress | Nitroprusiato de sodio;
- (CO) Colombia: Nitroprusiato de sodio | Prusiat;
- (CZ) Czech Republic: Nipride;
- (EC) Ecuador: Nitroprusiato de sodio;
- (GB) United Kingdom: Nitropress;
- (HK) Hong Kong: Nitropress;
- (HU) Hungary: Nipride;
- (JP) Japan: Nitopro;
- (KR) Korea, Republic of: Nipride;
- (LT) Lithuania: Nipride;
- (LU) Luxembourg: Nipruss;
- (LV) Latvia: Nipride;
- (MX) Mexico: Adprisat | Cordibux | Dorespin | Nitroprusiato de sodio;
- (MY) Malaysia: Nipride | Sodium nitroprusside;
- (NO) Norway: Nitropress | Nitroprusside sagent;
- (NZ) New Zealand: Sodium nitroprusside;
- (PE) Peru: Clenil | Nitropress;
- (PR) Puerto Rico: Nipride rtu | Nitropress | Sodium nitroprusside;
- (QA) Qatar: Nitropress | Nitroprussiat Fides;
- (RU) Russian Federation: Nanipruss;
- (SG) Singapore: DBL sodium nitroprusside;
- (SI) Slovenia: Nipruss | Nitropress;
- (TR) Turkey: Nipride | Nipruss;
- (UY) Uruguay: Niprusodio fada;
- (VE) Venezuela, Bolivarian Republic of: Nitroprusiato

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